

(DS Arrays)

✦ 1. Common Array Questions Asked in Interviews

◆ Basic Level

- Find maximum and minimum in an array
- Reverse an array
- Find the largest three elements
- Check if array is sorted
- Find the missing number in an array of 1 to N
- Find the duplicate in an array
- Rotate array by k steps (left/right)
- Move all zeros to the end
- Remove duplicates from sorted array
- Linear search / Binary search in array

◆ Intermediate Level

- Kadane's Algorithm (Maximum subarray sum)
- Trapping Rain Water problem
- Next permutation of array
- Two Sum / Three Sum / Four Sum
- Subarray sum equals k
- Longest consecutive sequence
- Maximum product subarray
- Find leaders in an array
- Equilibrium index (Prefix/Suffix sums)
- Stock Buy and Sell (I, II, with cooldown, with fee, at most k times)

◆ Advanced Level

- Merge Intervals
- Minimum number of platforms (interval overlap problem)
- Maximum circular subarray sum
- Sliding Window Maximum
- Median of two sorted arrays

- Search in rotated sorted array (with duplicates)
 - Product of array except self
 - Find kth smallest/largest element (QuickSelect / Heap)
 - Count inversions in an array (Merge Sort technique)
 - Maximum sum rectangle in a 2D matrix
 - Subarray with given XOR / sum
-

✦ 2. 150 LeetCode Array Questions (Categorized)

◆ Easy (Foundational)

◆ Basics

1. **Two Sum** (1)
2. **Best Time to Buy and Sell Stock** (121)
3. **Contains Duplicate** (217)
4. **Maximum Subarray** (53)
5. **Merge Sorted Array** (88)
6. **Remove Duplicates from Sorted Array** (26)
7. **Remove Element** (27)
8. **Plus One** (66)
9. **Valid Mountain Array** (941)
10. **Find Pivot Index** (724)

◆ Searching / Rotating

11. **Search Insert Position** (35)
12. **Squares of a Sorted Array** (977)
13. **Find Numbers with Even Number of Digits** (1295)
14. **Third Maximum Number** (414)
15. **Maximum Product of Two Elements in an Array** (1464)

◆ Rotation / Shifting

16. **Rotate Array** (189)
17. **Move Zeroes** (283)
18. **Build Array from Permutation** (1920)
19. **Concatenation of Array** (1929)
20. **Shuffle the Array** (1470)

◆ Prefix / Suffix

21. **Running Sum of 1d Array** (1480)

- 22. **Richest Customer Wealth** (1672)
- 23. **Minimum Value to Get Positive Step by Step Sum** (1413)
- 24. **Pivot Index** (724) (*classic prefix sum*)
- 25. **Find Pivot Index** (again different variants)

◆ Counting / Hashing

- 26. **Majority Element** (169)
- 27. **Single Number** (136)
- 28. **Intersection of Two Arrays** (349)
- 29. **Intersection of Two Arrays II** (350)
- 30. **Missing Number** (268)

◆ Simple Math / Tricks

- 31. **Max Consecutive Ones** (485)
- 32. **Max Consecutive Ones III (easy variant)** (1004 - edge medium)
- 33. **Find All Numbers Disappeared in an Array** (448)
- 34. **Find All Duplicates in an Array** (442 – borderline easy/medium)
- 35. **Monotonic Array** (896)

◆ 2D Arrays (Matrix)

- 36. **Reshape the Matrix** (566)
- 37. **Transpose Matrix** (867)
- 38. **Flipping an Image** (832)
- 39. **Diagonal Traverse** (498)
- 40. **Toeplitz Matrix** (766)

◆ Miscellaneous

- 41. **Lucky Numbers in a Matrix** (1380)
- 42. **Count Negative Numbers in a Sorted Matrix** (1351)
- 43. **Special Positions in a Binary Matrix** (1582)
- 44. **Check if N and Its Double Exist** (1346)
- 45. **Check if Array is Sorted and Rotated** (1752)
- 46. **Two Sum II – Input array is sorted** (167)
- 47. **Fair Candy Swap** (888)
- 48. **Best Time to Buy and Sell Stock II (easy variant)** (122)
- 49. **Kth Missing Positive Number** (1539)
- 50. **How Many Numbers Are Smaller Than the Current Number** (1365)

... (around 50 Easy array-focused questions)

◆ Medium (Core Interview Level)

16. 3Sum (15)
17. 4Sum (18)
18. Container With Most Water (11)
19. Subarray Sum Equals K (560)
20. Maximum Product Subarray (152)
21. Find Minimum in Rotated Sorted Array (153)
22. Search in Rotated Sorted Array (33)
23. Kth Largest Element in Array (215)
24. Product of Array Except Self (238)
25. Longest Consecutive Sequence (128)
26. Insert Interval (57)
27. Merge Intervals (56)
28. Minimum Size Subarray Sum (209)
29. Sliding Window Maximum (239)
30. Spiral Matrix (54)
31. Game of Life (289)
32. Set Matrix Zeroes (73)
33. Jump Game (55)
34. Jump Game II (45)
35. Gas Station (134)

◆ Subarrays / Prefix Sum

1. Subarray Sum Equals K (560)
 2. Maximum Product Subarray (152)
 3. Subarray Product Less Than K (713)
 4. Continuous Subarray Sum (523)
 5. Find Pivot Index (Harder variant with prefix) (724 - extension)
 6. Maximum Length of Subarray with Positive Product (1567)
 7. Longest Turbulent Subarray (978)
 8. Minimum Size Subarray Sum (209)
 9. Number of Sub-arrays of Size K and Average $\geq$ Threshold (1343)
 10. K Radius Subarray Averages (2090)
-

◆ Two Pointers / Sorting

11. 3Sum (15)
12. 3Sum Closest (16)

13. **4Sum** (18)
 14. **Sort Colors (Dutch National Flag)** (75)
 15. **Interval List Intersections** (986)
 16. **Boats to Save People** (881)
 17. **Two Sum II (Sorted Input)** (167 – medium extension)
 18. **Array With Elements Not Equal to Average of Neighbors** (1968)
 19. **Pairs of Songs With Total Durations Divisible by 60** (1010)
 20. **Reduce Array Size to The Half** (1338)
-

◆ **Sliding Window**

21. **Longest Substring Without Repeating Characters** (3 – string but array logic)
 22. **Fruit Into Baskets** (904)
 23. **Longest Repeating Character Replacement** (424 – string/array)
 24. **Sliding Window Maximum** (239)
 25. **Maximum Points You Can Obtain from Cards** (1423)
-

◆ **Greedy / Optimization**

26. **Jump Game** (55)
 27. **Jump Game II** (45)
 28. **Gas Station** (134)
 29. **Candy** (135 – greedy array)
 30. **Partition Labels** (763)
-

◆ **Matrix (2D Arrays)**

31. **Spiral Matrix** (54)
 32. **Spiral Matrix II** (59)
 33. **Set Matrix Zeroes** (73)
 34. **Rotate Image** (48)
 35. **Word Search** (79)
 36. **Search a 2D Matrix** (74)
 37. **Search a 2D Matrix II** (240)
 38. **Minimum Path Sum** (64 – DP grid)
 39. **Game of Life** (289)
 40. **Sort the Matrix Diagonally** (1329)
-

◆ **Advanced Array / Heap / Binary Search**

41. **Find First and Last Position of Element in Sorted Array** (34)

- 42. **Kth Largest Element in an Array** (215)
- 43. **Find Peak Element** (162)
- 44. **Search in Rotated Sorted Array** (33)
- 45. **Search in Rotated Sorted Array II** (81)
- 46. **Find Minimum in Rotated Sorted Array** (153)
- 47. **Find Minimum in Rotated Sorted Array II** (154)
- 48. **Product of Array Except Self** (238)
- 49. **Wiggle Subsequence** (376)
- 50. **Longest Consecutive Sequence** (128)

... (around 70 Medium array questions)

◆ **Hard (Challenging, FAANG-level)**

- 36. **Trapping Rain Water** (42)
- 37. **Median of Two Sorted Arrays** (4)
- 38. **First Missing Positive** (41)
- 39. **Maximum Gap** (164)
- 40. **Largest Rectangle in Histogram** (84)
- 41. **Maximal Rectangle** (85)
- 42. **Shortest Subarray with Sum $\geq K$** (862)
- 43. **Sliding Window Median** (480)
- 44. **Find the Duplicate Number** (287)
- 45. **Missing Number** (268)
- 46. **Count Inversions** (493 – Reverse Pairs)
- 47. **Candy** (135)
- 48. **N-Queens** (51 – uses arrays for placement)
- 49. **Maximum Sum of Rectangle No Larger Than K** (363)
- 50. **Regular Trapping Rain Water II** (407)

◆ **Prefix Sum / Subarray (Advanced)**

- 1. **Trapping Rain Water** (42)
 - 2. **Trapping Rain Water II** (407 – 2D version)
 - 3. **Maximum Sum of Rectangle No Larger Than K** (363)
 - 4. **Subarray Sum Equals K (Hard variant)** (862 – Shortest Subarray with Sum $\geq K$)
 - 5. **Count of Range Sum** (327)
 - 6. **Reverse Pairs (Count Inversions)** (493)
-

◆ Sliding Window / Deque / Heap

7. **Sliding Window Median** (480)
 8. **Median of Sliding Window II** (295 – Median Finder, often linked)
 9. **Longest Substring with At Most K Distinct Characters** (340 – string but array-based sliding window)
 10. **Minimum Window Subsequence** (727)
-

◆ Greedy / Optimization

11. **Candy (Hard distribution variant)** (135 – also seen in Medium, but tricky case handling)
 12. **Gas Station (Harder edge-case variants)** (134 extended)
 13. **Jump Game V** (1340)
 14. **Minimum Number of Refueling Stops** (871)
-

◆ Matrix (2D Arrays, Hard)

15. **Largest Rectangle in Histogram** (84)
 16. **Maximal Rectangle** (85)
 17. **Dungeon Game** (174 – grid DP, array-based)
 18. **Cherry Pickup** (741)
 19. **Cherry Pickup II** (1463)
 20. **Longest Increasing Path in a Matrix** (329)
-

◆ Binary Search / Advanced Array Logic

21. **Median of Two Sorted Arrays** (4)
 22. **Find K-th Smallest Pair Distance** (719)
 23. **Find K-th Smallest Sum of Two Sorted Arrays** (373)
 24. **Divide Chocolate / Split Array Largest Sum** (410)
 25. **Minimum Number of Days to Make m Bouquets** (1482)
-

◆ Miscellaneous (Hard Patterns)

26. **First Missing Positive** (41)
27. **Smallest Range Covering Elements from K Lists** (632)
28. **Super Egg Drop** (887 – DP with arrays)
29. **Maximum Gap** (164)
30. **Shortest Subarray with Sum at Least K** (862) (*critical sliding window hard*)

... (around 30 Hard array questions)

✓ Total = **150+ carefully selected LeetCode array questions** (mix of Easy, Medium, Hard)

These cover **all core patterns**:

- Prefix/Suffix Sum
- Sliding Window
- Two Pointers
- Binary Search on answer
- Hashing + Array
- Heap + Array
- DP on arrays

✦ 1. Common String Questions Asked in Interviews

◆ Basic Level

- Reverse a string
- Check if a string is palindrome
- Find first non-repeating character
- Count vowels/consonants
- Check if two strings are anagrams
- Remove duplicates from string
- Print all substrings of a string
- Check rotation of another string
- Implement strstr() / indexOf()

◆ Basics / Palindrome

1. **Valid Palindrome** (125)

2. **Valid Palindrome II** (680)
3. **Reverse String** (344)
4. **Reverse Vowels of a String** (345)
5. **Implement strStr()** (28)
6. **Length of Last Word** (58)
7. **Check if Word Equals Summation of Two Words** (1880)
8. **Add Strings** (415)
9. **To Lower Case** (709)
10. **Detect Capital** (520)

◆ **Anagrams / Counting**

11. **Valid Anagram** (242)
12. **Find the Difference** (389)
13. **Ransom Note** (383)
14. **First Unique Character in a String** (387)
15. **Isomorphic Strings** (205)
16. **Word Pattern** (290)
17. **Sort Characters By Frequency (easy variant)** (451 – borderline medium)
18. **Jewels and Stones** (771)
19. **Check if Sentence Is Pangram** (1832)
20. **Determine if Two Strings Are Close** (1657)

◆ **Simple Transformations**

21. **Excel Sheet Column Title** (168)
22. **Excel Sheet Column Number** (171)
23. **Convert Binary Number in a Linked List to Integer** (1290) (*string-ish*)
24. **Add Binary** (67)
25. **Defanging an IP Address** (1108)

◆ **Miscellaneous**

26. **Goat Latin** (824)
27. **Valid Word Abbreviation** (408)
28. **Check if a String Is a Prefix of Array** (1961)
29. **Check If Two String Arrays Are Equivalent** (1662)
30. **Capitalize the Title** (2129)
31. **Longest Common Prefix** (14)
32. **Robot Return to Origin** (657)
33. **Replace All Digits with Characters** (1844)
34. **Split a String in Balanced Strings** (1221)
35. **Count Binary Substrings** (696)
36. **Repeated Substring Pattern** (459)
37. **Valid Parentheses** (20)
38. **Balanced String Split** (1221)
39. **Roman to Integer** (13)
40. **Integer to Roman** (12)
41. **Remove All Adjacent Duplicates In String** (1047)

42. **Check If N and Its Double Exist (String variant)** (1346 adaptation)
43. **Check String Halves Alike** (1704)
44. **Make The String Great** (1544)
45. **Reverse Only Letters** (917)
46. **String Matching in an Array** (1408)
47. **Truncate Sentence** (1816)
48. **Check if a String Is Decomposable into Value-Equal Substrings** (1957)
49. **Day of the Week** (1185)
50. **Generate a String With Characters That Have Odd Counts** (1374)

◆ Intermediate Level

- Longest common prefix (among strings)
- Longest palindrome substring
- Longest substring without repeating characters
- Valid parentheses (balanced brackets)
- Implement atoi (string to integer)
- Group anagrams
- Word pattern matching
- String compression
- Minimum window substring containing all characters
- Check isomorphic strings

50 Medium String Questions (LeetCode)

◆ Substrings / Palindromes

1. **Longest Substring Without Repeating Characters** (3)
2. **Longest Palindromic Substring** (5)
3. **Palindromic Substrings** (647)
4. **Longest Palindromic Subsequence** (516)
5. **Count Palindromic Subsequences (basic DP)** (730)
6. **Substring with Concatenation of All Words** (30)
7. **Minimum Window Substring** (76)
8. **Longest Repeating Character Replacement** (424)
9. **Find All Anagrams in a String** (438)
10. **Permutation in String** (567)

◆ Anagrams / Grouping

11. **Group Anagrams** (49)
12. **Group Shifted Strings** (249)

- 13. **Valid Parenthesis String** (678)
- 14. **Custom Sort String** (791)
- 15. **Frequency Sort** (451)

◆ **Conversions / Parsing**

- 16. **String to Integer (atoi)** (8)
- 17. **Multiply Strings** (43)
- 18. **Integer to English Words** (273)
- 19. **Additive Number** (306)
- 20. **Basic Calculator II** (227)

◆ **DFS / Backtracking**

- 21. **Word Search** (79)
- 22. **Word Search II** (212)
- 23. **Restore IP Addresses** (93)
- 24. **Letter Combinations of a Phone Number** (17)
- 25. **Generate Parentheses** (22)

◆ **Sliding Window**

- 26. **Longest Substring with At Most Two Distinct Characters** (159)
- 27. **Longest Substring with At Most K Distinct Characters** (340)
- 28. **Minimum Window Subsequence** (727)
- 29. **Repeated DNA Sequences** (187)
- 30. **Smallest Subsequence of Distinct Characters** (1081)

◆ **DP / Advanced String Logic**

- 31. **Decode Ways** (91)
- 32. **Decode Ways II** (639)
- 33. **Word Break** (139)
- 34. **Word Break II** (140)
- 35. **Distinct Subsequences** (115)
- 36. **Edit Distance** (72)
- 37. **Scramble String** (87)
- 38. **Interleaving String** (97)
- 39. **Minimum ASCII Delete Sum for Two Strings** (712)
- 40. **Longest Common Subsequence** (1143)

◆ **Miscellaneous**

- 41. **Zigzag Conversion** (6)
- 42. **Count and Say** (38)
- 43. **Simplify Path** (71)
- 44. **Excel Column Titles Extended** (171, 168 – parsing)
- 45. **Check If a String Contains All Binary Codes of Size K** (1461)

- 46. **Reorganize String** (767)
- 47. **Partition Labels** (763)
- 48. **Reorder Data in Log Files** (937)
- 49. **Remove Invalid Parentheses** (301)
- 50. **Longest String Chain** (1048)

◆ **Advanced Level**

- Regular Expression Matching (Regex engine)
- Wildcard Matching
- Edit Distance (Levenshtein distance)
- Word Break problem
- Longest valid parentheses
- Count and Say sequence
- Decode Ways (dynamic programming)
- Z-algorithm / KMP (pattern searching)
- Rabin-Karp substring search
- Smallest window to make an anagram

◆ **Matching & DP**

1. **Regular Expression Matching** (10)
2. **Wildcard Matching** (44)
3. **Edit Distance (DP)** (72)
4. **Distinct Subsequences II** (940)
5. **Longest Duplicate Substring** (1044)
6. **Shortest Palindrome** (214)
7. **Word Ladder** (127)
8. **Word Ladder II** (126)
9. **Word Break III** (Hard extension of 139/140)
10. **Palindrome Partitioning II** (132)

◆ **Sliding Window / Subsequence**

11. **Minimum Window Subsequence** (727)
12. **Smallest Subsequence of Distinct Characters** (1081)
13. **Remove Duplicate Letters** (316)
14. **Longest Substring with At Most K Distinct Characters** (340 Hard version)
15. **Substring with Largest Variance** (2272)

◆ **Advanced String Search**

16. **Implement Trie (Prefix Tree)** (208)

- 17. **Design Add and Search Words Data Structure** (211)
- 18. **Stream of Characters** (1032)
- 19. **Word Search II** (212)
- 20. **Replace Words** (648)

◆ **Advanced DP / Graph + String**

- 21. **Alien Dictionary** (269)
- 22. **Reconstruct Itinerary** (332 – path as strings)
- 23. **Parse Lisp Expression** (736)
- 24. **Basic Calculator III** (772)
- 25. **Brace Expansion II** (1096)

◆ **Miscellaneous Hard**

- 26. **Binary Tree to String Conversion** (606 – hybrid)
 - 27. **Design Autocomplete System** (642)
 - 28. **Longest Valid Parentheses** (32 – string/stack/DP)
 - 29. **Wildcard Regular Expressions** (extension of 10 + 44)
 - 30. **Palindrome Pairs** (336)
-

✦ 2. 150 LeetCode String Questions (Categorized)

◆ Easy (Fundamentals)

1. Valid Palindrome (125)
 2. Implement strStr() (28)
 3. Length of Last Word (58)
 4. Add Binary (67)
 5. Valid Anagram (242)
 6. First Unique Character in String (387)
 7. Excel Sheet Column Number (171)
 8. Excel Sheet Column Title (168)
 9. Ransom Note (383)
 10. Isomorphic Strings (205)
 11. Detect Capital (520)
 12. Reverse String (344)
 13. Reverse Vowels of a String (345)
 14. Valid Parentheses (20)
 15. Longest Common Prefix (14)
 - ... (~50 easy string questions)
-

◆ Medium (Most Common in Interviews)

16. Longest Substring Without Repeating Characters (3)
17. Longest Palindromic Substring (5)
18. String to Integer (atoi) (8)
19. Integer to Roman (12)
20. Roman to Integer (13)
21. Group Anagrams (49)
22. Minimum Window Substring (76)
23. Decode Ways (91)
24. Simplify Path (71)
25. Word Pattern (290)
26. Count and Say (38)
27. Add Strings (415)
28. Multiply Strings (43)

29. Restore IP Addresses (93)
 30. Longest Repeating Character Replacement (424)
 31. Find All Anagrams in a String (438)
 32. Longest Palindromic Subsequence (516)
 33. Valid Palindrome II (680)
 34. Repeated DNA Sequences (187)
 35. Partition Labels (763)
 36. Palindromic Substrings (647)
 37. Substring with Concatenation of All Words (30)
 38. Remove Invalid Parentheses (301)
 39. Custom Sort String (791)
 40. String Compression (443)
 - ... (~70 medium questions)
-

◆ Hard (Challenging, FAANG-Level)

41. Regular Expression Matching (10)
42. Wildcard Matching (44)
43. Longest Valid Parentheses (32)
44. Edit Distance (72)
45. Distinct Subsequences (115)
46. Scramble String (87)
47. Word Ladder (127)
48. Word Ladder II (126)
49. Word Search II (212)
50. Minimum Window Subsequence (727)
51. Basic Calculator (224)
52. Basic Calculator II (227)
53. Basic Calculator III (772)
54. Smallest Subsequence of Distinct Characters (1081)
55. Longest Duplicate Substring (1044)
56. Regular Expression Matching with '.' and '*' (hard extension)
57. Reorganize String (767)
58. Check If a String Contains All Binary Codes of Size K (1461)

- 59. Remove Duplicate Letters (316)
 - 60. Shortest Palindrome (214)
 - ... (~30 hard questions)
-

✓ Together, these give you **150+ string questions** covering:

- Palindromes
- Substrings/Subsequences
- Anagrams
- Sliding Window on strings
- Pattern Matching (Regex, Wildcard, KMP, Z-function, Rabin-Karp)
- Dynamic Programming on strings (Edit distance, Decode ways, Word break)
- Stack-based problems (Valid parentheses, Calculator problems)